Team Control Number

16738

Problem Chosen

B

2025
HiMCM
Summary Sheet

Location Is Destiny: A Four-Stage MCDM Framework for Selecting Sustainable Mega-events Host Cities

Large-scale sport mega-events like the Super Bowl present a significant environmental challenge, with sustainability outcomes critically dependent on the selection of the host city. This paper moves beyond superficial operational fixes to address this core strategic issue by developing a comprehensive mathematical model for selecting the most environmentally sustainable host city. Our approach is centered on a novel **Four-Stage Assessment Framework** (Supply, Consumption, Generation, Capacity) that deconstructs an event's environmental footprint into a quantifiable, location-dependent process.

To operationalize this framework, we constructed a robust **Multi-Criteria Decision-Making (MCDM) model**. First, we established a system of twelve secondary indicators, quantifying each using data from authoritative public sources. We then developed a **Combined Weighting Model**, integrating the subjective Analytic Network Process (ANP) with the objective Entropy Weight Method to produce a balanced and scientifically grounded set of indicator weights. The final sustainability ranking for 22 candidate cities was then calculated using the **TOPSIS method**. Our baseline model demonstrated high validity, aligning with real-world NFL decisions by ranking the San Francisco Bay Area (host of Super Bowl LX) as the second most sustainable option.

To provide a forward-looking recommendation for the 2029 Super Bowl, we incorporated a **Trend Extrapolation Correction Model**. By projecting policy-driven changes in key indicators like Grid Cleanliness and Waste Management, we forecasted the 2029 sustainability landscape. The analysis revealed distinct "archetypes" of sustainability—from the inherently green "All-Rounder" to the policy-driven "Improver." After applying a real-world "cooling-off period" constraint, our model recommends **Seattle, WA**, as the premier choice among new potential hosts due to its outstanding all-around performance, and the **Los Angeles Area** as the best option among experienced hosts, driven by its aggressive clean energy policies.

Finally, the model was extended to evaluate multi-venue mega-events like the Olympics, demonstrating its versatility. Our work provides the NFL with a transparent, data-driven, and strategically insightful tool for making future host city selections that are genuinely aligned with long-term environmental goals.

Keywords: Sport Mega-Events; Sustainability; Host City Selection; Multi-Criteria Decision-Making (MCDM); Four-Stage Framework; Analytic Network Process (ANP); TOPSIS

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1 Introduction and Problem Restatement

1.1 Background

Large-scale sport mega-events like the Olympic Games and the Super Bowl are defining features of our global culture, generating substantial socio-economic benefits. However, these events also leave a significant environmental footprint. Local ecosystems suffer immense pressure from the intensive resource consumption, large-scale construction, and concentrated global travel required for these events. Although sustainability has become a key focus for event organizers due to the rapid growth of global environmental awareness, a considerable gap often persists between sustainability commitments and actual environmental outcomes, highlighting the need for more scientific and transparent planning tools.

The NFL's Super Bowl provides a perfect case study for this challenge. As a single-venue event that moves to a new location annually, it allows us to isolate and study a critical, yet often underestimated, variable in the sustainability equation: **the host city**. An event's environmental impact is not a fixed attribute; the specific context of its location fundamentally shapes it. The carbon intensity of the stadium's electricity, the efficiency of public transportation for fans, and the capacity of local water and waste systems are all determined by the host city's unique geographical and infrastructural characteristics.

This leads to a key insight: achieving true sustainability requires a shift in perspective. Instead of relying on temporary fixes, such as adding solar panels to a stadium in a city powered by a coal-dominant grid, we need a more systematic approach to sustainability. The most impactful decision in making an event sustainable is choosing a host city that already possesses a robust and resilient environmental infrastructure. The challenge, therefore, is not just to make the event *look* green, but to select a location where the event can seamlessly integrate with an already sustainable urban system. This paper aims to develop a mathematical model that places this environmental logic at the very core of the host city selection process.

1.2 Our Approach: A Multi-Criteria Decision-Making (MCDM) model

To address the problem as tasked by the Consulting Organization Making Athletics Planet positive (COMAP), our work is structured as a comprehensive, multi-layered investigation. We will develop a model to evaluate and provide insights into sustainable event hosting.

First, we will deconstruct the complex environmental footprint of a mega-event into a structured framework of quantifiable indicators. Using Super Bowl LIX in New Orleans as a guiding case study, we will identify the key drivers of environmental impact, from resource supply and consumption to waste generation and the local environment's capacity to absorb these pressures. This allows us to understand *how* and *why* the specific characteristics of a host city matter.

Based on these indicators, we will construct a quantitative **Multi-Criteria Decision-Making (MCDM) model** to rank potential host cities based solely on environmental performance. To ensure a robust and objective evaluation, we will integrate two powerful techniques: a combined weighting model (**Analytic Network Process (ANP)** and **Entropy Weight Method**) to determine the weights of our indicators by capturing their complex interdependencies, and the **TOPSIS method** to calculate a final sustainability score for each city.

With our model built and validated, we will apply it to recommend the most sustainable host for the 2029 Super Bowl, considering both cities with prior hosting experience and those new to the event. To showcase the model's versatility, we will then extend its logic to evaluate larger, multi-venue competitions, such as the Olympics, incorporating additional factors, including inter-city travel and the impact of new construction. This entire process will culminate in a formal recommendation letter to the NFL, detailing our findings and the strategic rationale behind our choice.

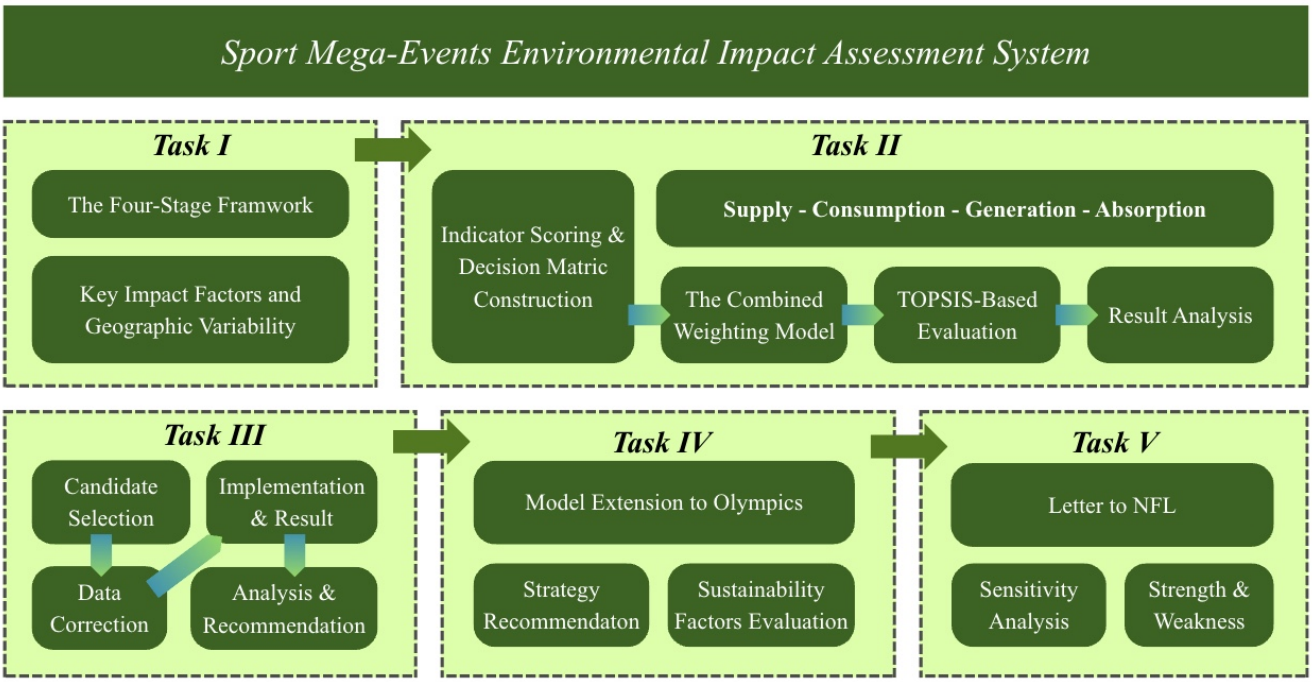


Figure 1.1: The overall structure of our work

2 Assumptions and Justifications

- **Assumption 1:** We assume that the on-site resource consumption (e.g., water usage, waste generation) and behavior patterns of an average attendee are consistent across different host cities for a Super Bowl event.

  Justification: This assumption allows our model to focus on the structural and geographical differences between cities. A mega-event like the Super Bowl operates within a highly standardized and controlled environment, thereby official vendors, operational protocols, and the nature of the fan experience are largely uniform, leading to consistent per-capita consumption and waste baselines.

- **Assumption 2:** We assume that the nationally representative distribution based on the fanbases of all NFL teams can present the geographical distribution of traveling attendees.

  Justification: It is impossible to predict the exact travel patterns for all traveling attendees. By assuming a composite, national fan distribution, we can calculate a stable and comparable "average travel footprint" for each potential host city.

- **Assumption 3:** We assume that a city's core infrastructure remains stable, such as the public transit network, airport location, and the carbon intensity of its electricity grid.

  Justification: A week-long event, despite its intensity, has a negligible short-term impact on a city's macro-level infrastructure or its regional energy generation mix.

- **Assumption 4:** We assume that the selected environmental indicators within each category are sufficiently independent to be evaluated separately.

  Justification: To reduce multi-collinearity in the ecosystem, we select indicators representing distinct environmental dimensions (e.g., air quality is assumed not to directly affect water stress). ANP is then used to partially capture interdependencies among criteria.

- **Assumption 5:** We assume that data from public sources (e.g., government databases, official reports, and established scientific indices) are accurate and representative of the conditions in the host cities.

  Justification: By relying on reputable and standardized data sources, we ensure that our analysis is as objective and replicable as possible.

3 Symbol

Table 3.1: Symbols and Notations Used in the Models

Symbol	Description	Symbol	Description
A	Area (m^2)	M	Waste and Material Score
b	Influence coefficient (-)	MF	Maintenance Factor (-)
C	Carbon emissions (t CO_2)	N	Number of people (people)
c	Carbon intensity (kg CO_2/MWh)	P	Power (MW)
$CCES$	City Clean Energy Score (-)	p	Proportion/Probability (-)
$CRSC$	Clean Resource Supply Capacity (-)	r	Rate/Ratio (-)
D	Distance in TOPSIS (-)	S	Overall Sustainability Score
D_1	Number of irrigation days (-)	T	Temperature (°C)
d	Distance (km)	t	Time duration (hours)
E	Energy consumption (MWh)	w	Weight vector (-)
EF	Emission factor (g CO_2/MWh)	X, Z, V	Matrices for TOPSIS (-)
$ELBC$	Environmental Load-Bearing Capacity (-)	α	Renewable energy ratio (-)
f	Efficiency factor (-)	λ	Weight tuning coefficient (-)
g	Goal achievement coefficient (-)		
k	Coefficient/Factor (-)		
$LEED$	LEED certification factor (-)		

4 Task 1: Deconstructing Environmental Impact via Four-Stages

The National Football League (NFL) has formally integrated sustainability into its strategic vision, not only as a member of the Green Sports Alliance but also through its dedicated NFL Green initiative [**1**]. This program is designed to mitigate the environmental impact of the NFL's major events and cultivate a lasting green legacy within each host community [**2**]. For the Super Bowl LIX in New Orleans, these commitments are translating into tangible actions. For instance, the host committee is enhancing waste management by significantly increasing recycling infrastructure and improving sorting processes to maximize material recovery [**3**]. Concurrently, resource circularity is being promoted through partnerships with local nonprofits, such as the Second Harvest Food Bank, to ensure surplus food is redistributed to address community needs.

4.1 A Four-Stage Framework for Environmental Impact Assessment

To systematically analyze the location-dependent variability of a mega-event's environmental footprint, we conceptualize its impact as a linear process that unfolds across four distinct stages: **Supply**, **Consumption**, **Generation**, and **Absorption**. This framework allows us to deconstruct the complex web of interactions between the event and its host city, moving from the city's inherent characteristics to the event's direct outputs and the environment's ability to respond. Each stage is defined as follows:

- **Stage 1: Clean Resource Supply.** This initial stage assesses the host city's foundational infrastructure and resource base. It evaluates the inherent "greenness" of the resources the event will draw upon, such as the local electricity grid's carbon intensity and the efficiency of its public transportation network. A city with a cleaner supply base provides a more sustainable starting point, regardless of the event's specific operations.
- **Stage 2: Operational Resource Consumption.** This stage quantifies the direct demand the event places on local resources, a demand that is heavily influenced by the host city's specific geographical and climatic context. It focuses on how local conditions, like climate, drive the consumption of

resources such as energy for heating or cooling, and how factors like regional water scarcity amplify the environmental cost of water usage.

- **Stage 3: Event-Generated Load.** This stage measures the direct environmental outputs—emissions and waste—produced by the event's activities. It accounts for the primary pollution sources, including the massive carbon footprint from spectator travel and the significant volume of solid waste generated on-site. The scale of this load is determined by both the event's nature and the host city's characteristics.
- **Stage 4: Environmental Load-Bearing Capacity.** The final stage evaluates the ability of the host city's natural and built environment to absorb, mitigate, and recover from the loads generated in the previous stage. This includes the ecosystem's resilience, measured by factors like baseline air quality, the presence of green spaces, and the region's long-term vulnerability to environmental shocks like droughts.

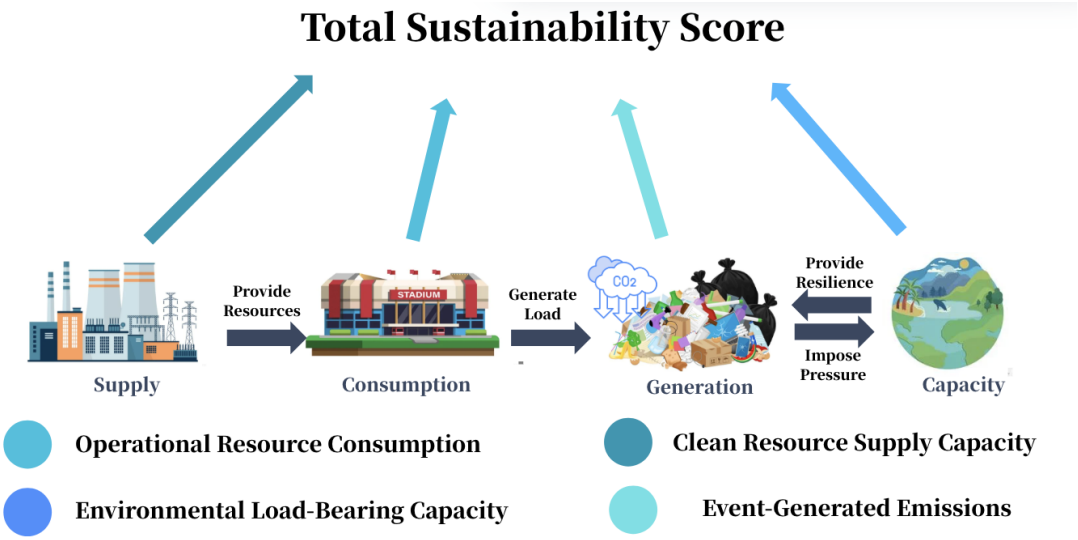


Figure 4.1: The Four-Stage Linear Process Framework for Mega-Event Environmental Impact Assessment.

As illustrated in Figure 7.2, this process-oriented approach provides a logical foundation for identifying key performance indicators at each stage. It forms the backbone of the quantitative evaluation model developed in the subsequent chapters by ensuring that our analysis is both comprehensive and systematic.

4.2 Identifying Key Impact Factors and Their Geographic Variability

Using the four-stage framework as our analytical lens, we now investigate the key factors that define an event's true environmental footprint at each stage. Our analysis focuses on identifying factors that are not only significant in their impact but are also highly sensitive to the unique characteristics of the host city, thereby justifying a location-centric evaluation model.

Stage 1: Clean Resource Supply

This initial stage assesses the host city's inherent environmental "starting line." It answers the critical question: *How "green" is the foundational infrastructure that the event will rely upon?* A city with a cleaner resource base will inherently host a more sustainable event, regardless of on-site operational efforts.

The sustainability of an event's foundation is determined by the city's existing infrastructure. A primary factor is **Energy Cleanliness**, which refers to the carbon intensity of the regional energy structure. An event's massive energy demand means that the local carbon footprint is directly inherited. This creates a stark geographical variance. For example, hosting in a city like **Seattle** (Washington), which benefits from extensive hydroelectric power, results in inherently lower emissions than hosting in a city like **Indianapolis**

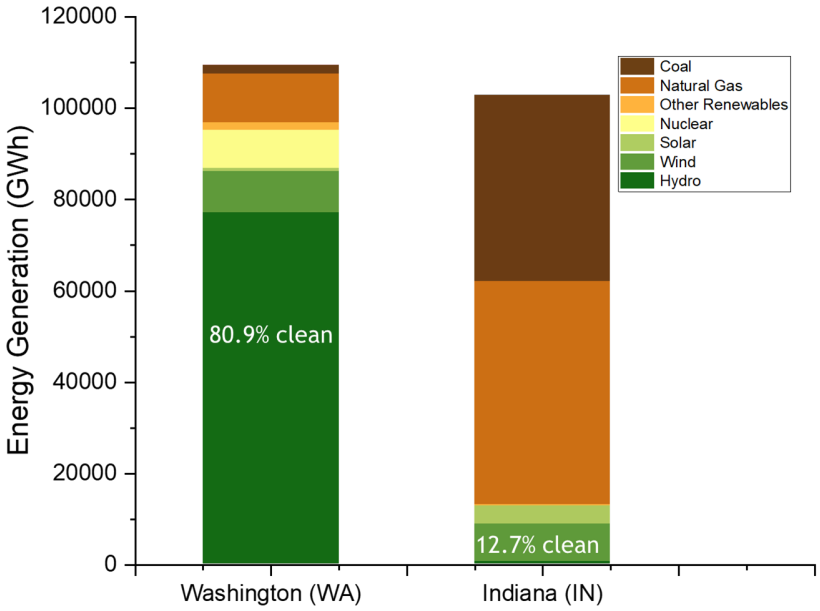


Figure 4.2: Electricity Generation Mix by Source for Washington and Indiana (2023). Washington's grid, powering Seattle, is predominantly renewable, whereas Indiana's grid, powering Indianapolis, is fossil-fuel dependent. Data Source: U.S. Energy Information Administration **5**.

(Indiana), located in a region with a historically coal-heavy energy mix. As visualized in Figure 5.2, Washington's power generation is dominated by renewables, while Indiana's relies heavily on fossil fuels, making Seattle a far superior choice in this regard.

Table 4.1: Public Transit Score Comparison for Selected Cities

City	Public Transit Score (0-100)
New York, NY	84.4
San Francisco, CA	77.4
Los Angeles, CA	50.9
Nashville, TN	**23.3**

*Source: Walk Score® (walkscore.com, 2024)***[6]**

In addition, **Public Transit Infrastructure** is another critical factor. A robust network of subways, light rail, and buses is the most effective tool for managing the movement of hundreds of thousands of attendees, mitigating traffic congestion and associated emissions. The difference between cities is profound. For example, a city like **New York**, with its comprehensive subway system, has a far greater capacity to absorb local travel demand sustainably than a more car-dependent city like **Nashville**. The disparity is quantified by the Public Transit Score, where New York vastly outperforms Nashville, as shown in Table 4.1.

Stage 2: Operational Resource Consumption

This stage quantifies the direct demand the event places on a city's resources, a demand heavily influenced by local conditions. It addresses the question: *Given the city's specific environment, how intensive will the event's resource consumption be?*

Table 4.2: 30-Year Average Climate Load Data for February (1991-2020)

City	Heating Degree Days (HDD)	Cooling Degree Days (CDD)	Total Climate Load
Minneapolis, MN	1332.3	0.0	1332.3
Miami, FL	22.8	194.5	217.3
San Diego, CA	288.7	0.0	288.7

Source: NOAA National Centers for Environmental Information (NCEI)[7]
Note: Degree Days are in Fahrenheit-days.

The local climate determines the baseline energy required for heating, ventilation, and air conditioning (HVAC). This factor creates a clear geographical trade-off. A February Super Bowl in a cold-climate city like **Minneapolis** would necessitate a massive energy expenditure for heating, as measured by its high Heating Degree Days (HDD). Conversely, hosting in a warm-climate city like **Miami** would require a similarly enormous energy load for air conditioning, reflected in its high Cooling Degree Days (CDD). Table 4.2 starkly illustrates these opposing climatic extremes. A temperate location like San Diego serves as a baseline, demonstrating a significantly lower combined energy demand.

Table 4.3: Baseline Water Stress Comparison for Selected Cities

City	Risk Classification	Score (0-5 scale)
Las Vegas, NV	Extremely High	4.0 - 5.0
Phoenix, AZ	Extremely High	4.0 - 5.0
Chicago, IL	Low	0.0 - 1.0

Source: World Resources Institute (WRI) Aqueduct Water Risk Atlas 4.0[8]

In addition, the water consumption associated with a large sporting event imposes varying levels of stress on different host regions, depending on their underlying hydrological conditions. This pressure can be quantitatively evaluated using the **Baseline Water Stress** index, which measures the ratio of total water withdrawals to renewable water supply. Because this indicator is highly sensitive to local climate factors such as precipitation levels and evaporation rates, the environmental impact of hosting an event becomes strongly location-dependent. For instance, staging an event in arid metropolitan areas such as **Las Vegas** or **Phoenix** poses substantially greater sustainability risks than in water-abundant cities like **Chicago**. As shown in Table 4.3, data from the World Resources Institute classify both Las Vegas and Phoenix as facing "Extremely High" water stress, highlighting the severe conservation challenges in these regions.

Stage 3: Event-Generated Load

This stage focuses on the direct environmental outputs—emissions and waste—produced by the event's activities. It answers the question: *What is the scale of the pollution and waste burden that the event will create?*

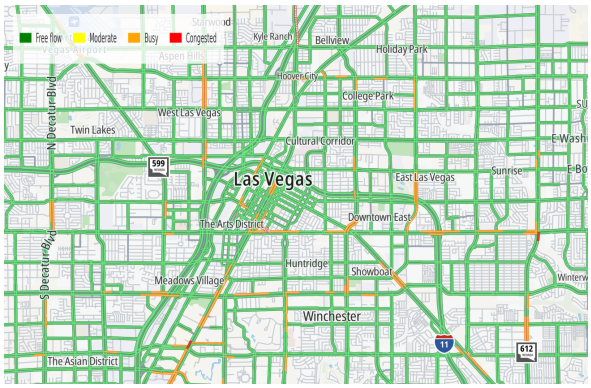


Figure 4.3: Las Vegas transportation map

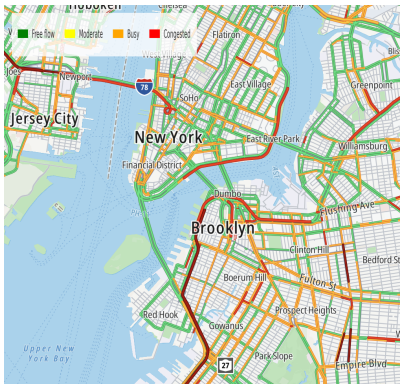


Figure 4.4: New York City transportation map

In the Las Vegas map, almost all the streets are green, which shows that the city experiences little congestion and that traffic flows freely. In contrast, in the New York–Brooklyn map, many major roads and bridges are yellow, orange, or bright red—especially around Lower Manhattan and the East River crossings—revealing much heavier traffic and frequent congestion.

Stage 4: Environmental Capacity

The final stage evaluates the host city's ecosystem's ability to buffer and absorb the impacts generated in the previous stage. It addresses the crucial insight that the same amount of pollution has different consequences in different environments. The key question is: *How resilient is the local environment to the additional stress of the event?*

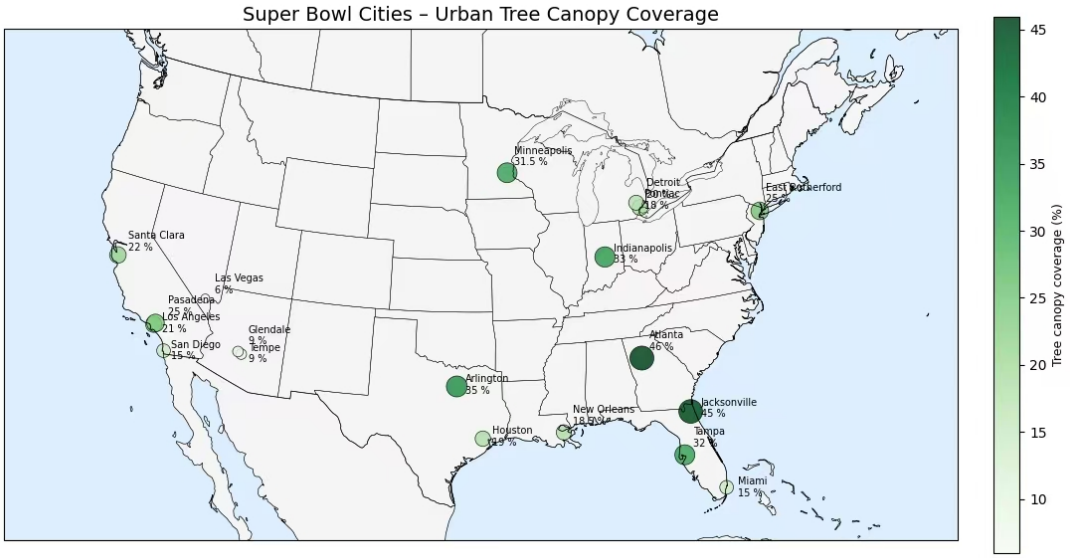


Figure 4.5: Urban Tree Canopy Coverage for U.S. Super Bowl Host Cities. The map illustrates the significant variation in green infrastructure, a key component of a city's environmental capacity. Data Source: U.S. Forest Service, "Urban Tree Canopy Cover in the United States" [**9**].

For example, **Urban Ecological Space**, often measured by urban tree canopy coverage, is critical for ecosystem services. Trees absorb CO, filter air pollutants, reduce stormwater runoff, and mitigate the urban heat island effect, acting as a city's natural defense system. As visualized in Figure 4.5, there is a vast difference in the green infrastructure among potential host cities. Cities in the Southeast like **Atlanta** (46%) and **Jacksonville** (45%) possess a dense tree canopy, providing a strong natural buffer. In stark contrast, desert cities like **Las Vegas** (6%) and **Phoenix** (Glendale, 9%) have minimal tree coverage, indicating a much lower natural capacity to mitigate environmental pressures.

5 Task 2: Construction of the Sustainability Evaluation Model

5.1 Model Overview

To quantitatively evaluate and rank potential Super Bowl host cities based entirely on environmental criteria, we developed a comprehensive Multi-Criteria Decision-Making (MCDM) model. Our approach is structured as a three-step process, as illustrated in Figure 7.2 Task II, which ensures a transparent, robust, and replicable evaluation.

- **Step 1: Indicator Scoring and Decision Matrix Construction.** We first process raw data for each of our secondary indicators, converting them into a standardized 0-10 score. This results in a decision matrix where each city's performance on every indicator is clearly quantified.

- **Step 2: Combined Weighting Model.** We then determine the relative importance (weight) of each indicator. To balance expert judgment with objective data trends, we employ a novel combined weighting scheme, utilizing both the Analytic Network Process (ANP) and the Entropy Weight Method.
- **Step 3: TOPSIS-Based Comprehensive Evaluation.** Finally, we synthesize the indicator scores and their corresponding weights using the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method. This calculates a final, holistic sustainability score for each city, allowing for a definitive ranking.

5.2 Step 1: Indicator Scoring and Decision Matrix Construction

The foundation of our model is the conversion of diverse raw data into a standardized and comparable format. Our twelve secondary indicators are quantified using data from reputable public sources to ensure objectivity and replicability. All scores are normalized to a 0-10 scale, where 10 represents the best performance.

5.2.1 Supply Indicators

We constructed a composite index based on three metrics: public transit efficiency, clean energy ratio, and energy utilization efficiency.

Public Transit Efficiency reflects a city's transportation conditions. This indicator integrates multiple factors, including urban transport infrastructure, traffic management, and residents' travel behavior.

Clearn Energy Ratio reflects the carbon intensity of the city's energy supply, indicating the structural potential for low-carbon operations. We collected the energy consumption data of the target cities and calculated their clean energy ratios. Specifically, we consider six categories of energy consumption: *oil*, *natural gas*, *coal*, *nuclear*, and *other renewable energy*. The data is shown in the figure below:

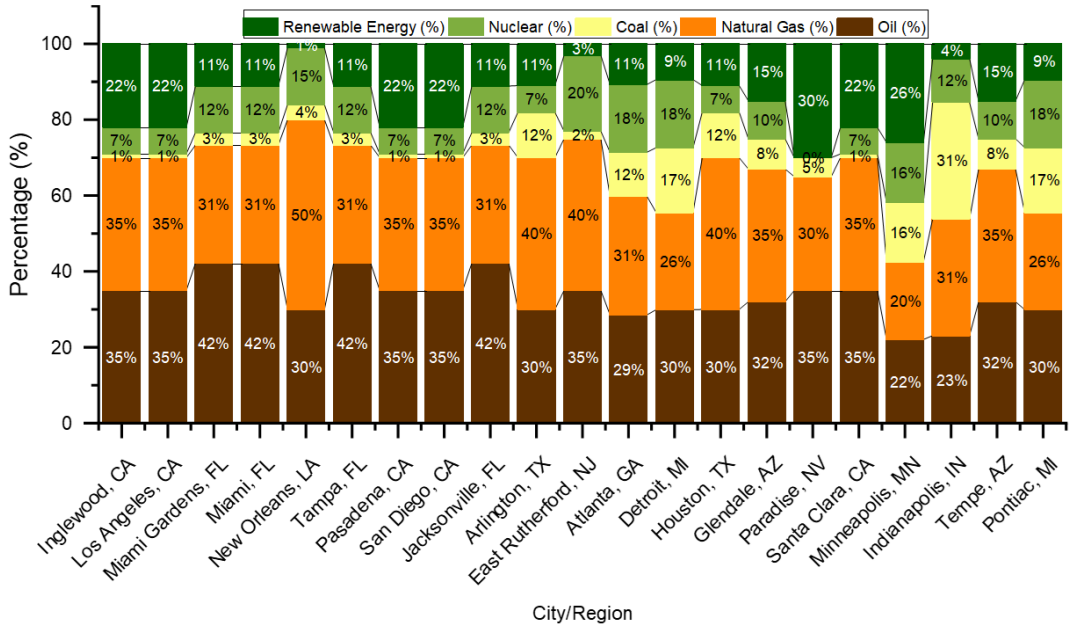


Figure 5.1: Energy structure for each city

The clean energy ratio is computed as

$$\text{Clean Energy Ratio} = \frac{E_{\text{nuc}} + E_{\text{ren}}}{E_{\text{oil}} + E_{\text{gas}} + E_{\text{coal}} + E_{\text{nuc}} + E_{\text{ren}}}. \tag{5.1}$$

Energy Utilization Efficiency captures how effectively the city converts energy input into functional services. Cities with modernized grids, efficient building stock, and advanced energy management systems

operate with inherently lower environmental footprints. When a mega-event adds significant electricity demand, a high-efficiency system prevents disproportionate increases in emissions.

5.2.2 Operational Resource Consumption

Water Usage. The total water consumption of a stadium consists of two components: audience water consumption and turf irrigation water consumption. We use the following formulas for the calculation:

$$W_{\text{specific}} = \alpha \cdot U_{\text{participant}} \cdot \beta + \gamma \cdot S_{\text{operation}} \cdot \delta \tag{5.2}$$

where α is the water consumption per participant per unit time (e.g., per person per day), $U_{\text{participant}}$ is the number of participants, and β is an adjustment factor for different event consumption patterns. γ is the water consumption per square meter of venue per day for operational maintenance (e.g., grass watering, cleaning), $S_{\text{operation}}$ is the venue area (m^2), and δ adjusts for the intensity of water usage based on event requirements.

Energy Consumption. The energy consumption of each stadium varies based on factors such as its local climate, the use of renewable energy, and the building's overall environmental performance.The model for estimating the total electricity consumption of a venue is expressed by the equation:

$$E_{\text{venue}} = P_{\text{base}} \times t \times b_{\text{temperature}} \times f_{\text{LEED}} \tag{5.3}$$

Where the parameters are defined as: E_{venue} = Total electricity consumption of the venue (MWh), P_{base} = Base power (MW), fixed at 12 MW (average for NFL events) , $B_{\text{temperature}}$ = Temperature influence coefficient, and f_{LEED} = LEED Energy Efficiency Factor (official measurement)

The temperature influence coefficient is calculated by following equaltion:

$$B_{\text{temperature}} = 1 + k \cdot |T_{\text{game day}} - 22| \tag{5.4}$$

where $|T_{\text{game day}} - 22|$ is the absolute difference between the actual temperature and the target indoor temperature (22°C).And k is the sensitivity coefficient, adjusting how much the temperature difference affects energy consumption.

5.2.3 Event-Generated Emissions and Waste

Carbon Emission. We develop an equation to quantify the carbon emissions generated by local travel. By categorizing transportation into public and private modes, we can accurately calculate the total carbon emissions produced locally during Super Bowl events.

$$C_{\text{travel}} = 2 \times d \times N \times [p \times EF_{\text{public}} + (1 - p) \times EF_{\text{private}}] \tag{5.5}$$

Where C is the total carbon emissions (tons of CO_2) from the event; d is the distance from the airport to the stadium, affecting travel emissions; N is the number of attendees; and p is the proportion of attendees using public transportation (0–1). The terms EF_{public} and EF_{private} are the per-person, per-kilometer carbon emission factors for public transport and private cars, respectively (g CO_2/km).

Electricity. Electricity generated from non-renewable sources contributes significantly to carbon emissions. Therefore, based on the share of renewable energy used by each stadium and the carbon intensity of the local grid (measured in CO_2 per TWh), we establish a formula to quantify the carbon emissions associated with electricity consumption.

$$C_{\text{electric}} = E_{\text{venue}} \times (c_{\text{electricity usage}}) \tag{5.6}$$

In specific, we break down the factor $c_{\text{electricity usage}}$ as follow:

$$c_{\text{electricity}} = (1 - \alpha) \cdot c_{\text{grid}}, \tag{5.7}$$

Where the parameters are defined as: α = Proportion of the stadium's electricity consumption supplied by on-site renewable energy systems. c_{grid} = Carbon intensity of the local electricity grid, measured as CO_2 emissions per TWh.

Waste and Material. Waste generated at the Super Bowl—from catering, giveaways, and temporary construction—is modeled using the **Waste and Material Sustainability Score** (M_s). To simplify, M_s depends on two key factors: the event's waste diversion rate (r_c) and the total number of attendees (N), assuming constant per-person waste:

$$M_s = 10 - \left(10 \times (1 - r_c) + k \cdot \frac{N}{N_0}\right) \tag{5.8}$$

Here, M_s ranges from 0 to 10, with higher values indicating better sustainability. The coefficient k captures scale effects (suggested range: 0.5–2.0), and N_0 is a reference population for normalization. This score provides a simplified but effective measure of the environmental impact of large events.

5.2.4 Environmental Load Capacity

This model quantifies a city's inherent ability to absorb the additional environmental strain from a mega-event. A higher ELBC score indicates a more resilient urban ecosystem. The score for city i is normalized to a 0–10 scale. The composite capacity coefficient α_i is calculated by integrating three key normalized (0–1) indicators: the **Tree Canopy Cover Rate** $p_{\text{Tree},i}$; the **Water Stress Index** $p_{\text{WSI},i}$ (a higher score indicates more water availability); the **Water Quality Index** p_{WQI} and the **Air Quality Index** $p_{\text{AQI},i}$ (a higher score indicates cleaner air). The weight of each factor is calculated by the Combined Weighting Model.

$$\alpha_i = w_{\text{Tree}}\, p_{\text{Tree},i} + w_{\text{WSI}}\, p_{\text{WSI},i} + w_{\text{WQI}}\, p_{\text{WQI},i} + w_{\text{AQI}}\, p_{\text{AQI},i} \tag{5.9}$$

5.2.5 Data Processing

All raw data for these indicators is then normalized to a 0-10 score to construct the final Decision Matrix. Positive indicators are normalized using the formula:

$$\text{Score} = 10 \times \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \tag{5.10}$$

Negative indicators are normalized using the inverted formula:

$$\text{Score} = 10 \times \left(1 - \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}\right) \tag{5.11}$$

where x_i is the raw value for city i, and $x_{\max}$ and $x_{\min}$ are the maximum and minimum values across all candidate cities for that indicator.

5.3 Step 2: The Combined Weighting Model

The determination of indicator weights is a critical step in any MCDM model, as it defines the relative importance of each criterion in the final evaluation. To create a balanced and robust weighting scheme that mitigates the biases of any single method, we combine a subjective approach (Analytic Network Process) with an objective approach (Entropy Weight Method).

5.3.1 Subjective Weighting: Analytic Network Process (ANP)

To capture expert opinions on strategic priorities and the complex interdependencies between criteria, we employ the ANP. Unlike the simpler AHP, ANP allows us to model a network of influences, which is suitable for our four-stage framework where each stage can influence the others.

Step 1: Construct the Network Structure and Judgment Matrices. We first established the network structure, defining the relationships between our four primary indicators (Supply, Consumption, Load, Capacity). Subsequently, simulating an expert panel, we performed pairwise comparisons using Saaty's 1-9 scale to quantify the relative importance between connected criteria. This process generated the necessary judgment matrices. The following table shows how we make the comparisons:

a_{ij}	a_{ji}	Interpretation
1	1	Criterion i and j are equally important
3	$\frac{1}{3}$	Criterion i is slightly more important than j
5	$\frac{1}{5}$	Criterion i is moderately more important than j
7	$\frac{1}{7}$	Criterion i is strongly more important than j
9	$\frac{1}{9}$	Criterion i is extremely more important than j

The following graph and tables show the comparisons of indicators

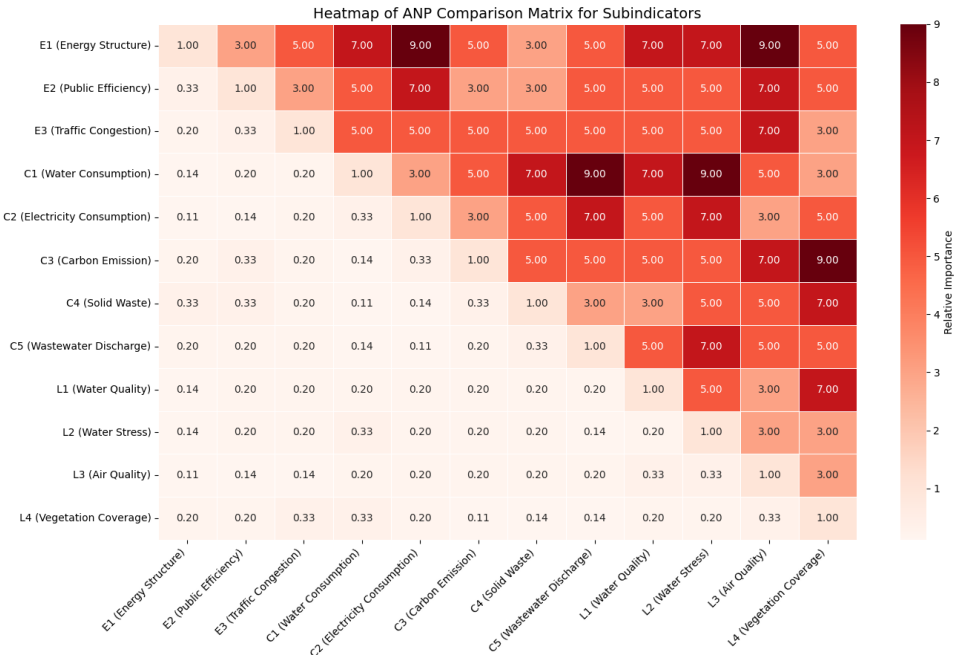


Figure 5.2: Relative importance heatmap

Step 2: Compute the Supermatrix and Final ANP Weights. The pairwise comparisons are used to form an unweighted supermatrix, which represents the initial influence levels. This is then stochasticized to create the weighted supermatrix, $\bar{W}$. To capture the total long-term influences, we calculate the limit supermatrix by raising the weighted supermatrix to a large power until it converges:

$$\bar{W}^{\infty} = \lim_{k \to \infty} \bar{W}^{k} \tag{5.12}$$

The final ANP weight vector, W_{ANP}, is extracted from the converged limit supermatrix.

A matrix $A = (a_{ij})$ must satisfy the consistency condition, checked by the Consistency Ratio (CR):

$$CR = \frac{CI}{RI} = \frac{(\lambda_{\max} - n)/(n-1)}{RI} \tag{5.13}$$

where $\lambda_{\max}$ is the principal eigenvalue of the matrix and RI is the Random Consistency Index.

After processing the judgment matrices and calculating the limit supermatrix, the resulting ANP weight vector was found in Table 5.1. The overall consistency ratio for the model was calculated as $CR < 0.1$, indicating that our judgments are acceptably consistent.

Table 5.1: Normalized Indicator Weights of ANP (Two-Column Layout).

Indicator	Weight	Indicator	Weight
Urban Energy Structure	0.13	Event Electricity Consumption	0.10
Urban Public Energy Efficiency	0.09	Event Carbon Emissions (Venues, Transportation)	0.15
Urban Traffic Congestion Coefficient	0.07	Event Solid Waste	0.04
Event Water Consumption	0.11	Event Wastewater Discharge	0.04
Urban Water Quality	0.07	Urban Air Quality	0.09
Urban Water Stress	0.05	Urban Vegetation Coverage	0.06

5.3.2 Objective Weighting: Entropy Weight Method

To incorporate the information inherent in the data itself, we use the Entropy Weight Method. This method assigns higher weights to indicators that exhibit greater variation across the candidate cities, as they provide more information for differentiation. The process uses the Decision Matrix $X = (x_{ij})$ from Step 1 as input.

Step 1: Normalize and Standardize the Data. First, each element x_{ij} in the decision matrix is normalized to calculate its proportion p_{ij} within its indicator column:

$$p_{ij} = \frac{x_{ij}}{\sum_{i=1}^{m} x_{ij}} \tag{5.14}$$

where m is the number of cities.

Step 2: Calculate Information Entropy. The information entropy E_j for each indicator j is calculated, which measures the degree of uncertainty or variation in the data:

$$E_j = -k \sum_{i=1}^{m} p_{ij} \ln(p_{ij}), \quad \text{where } k = \frac{1}{\ln(m)} \tag{5.15}$$

Step 3: Calculate Objective Weights. The final entropy weight w_j for each indicator is derived from its information entropy. A lower entropy (less uncertainty, more variation) results in a higher weight.

$$w_j = \frac{1 - E_j}{\sum_{j=1}^{n}(1 - E_j)} \tag{5.16}$$

This process yields the objective weight vector, $W_{Entropy}$.

$$W_{Entropy} = [0.18, 0.08, 0.06, 0.11, 0.13, 0.22, 0.09, 0.03, 0.05, 0.02, 0.01, 0.02]$$

Notably, 'Travel Footprint' received the highest objective weight, indicating that the candidate cities exhibit the greatest variation on this metric.

5.3.3 Sigmoid Function for Dealing with Extreme Data Value

If some values reach an extreme value, the weights we have calculated before will not describe the whole story. We designated a threshold value that indicate the shift between constant weights and dynamic weights. This smooth dynamic adjustment mechanism is processed using a Sigmoid function. The following pairwise function shows how it works:

$$\lambda(x) = \begin{cases} \lambda_{\text{base}}, & x < T_{\text{start}} \\ \lambda_{\text{base}} + (\lambda_{\max} - \lambda_{\text{base}}) \cdot \dfrac{1}{1 + e^{-k(x - T_{\text{trigger}})}}, & T_{\text{start}} \le x \le T_{\text{end}} \\ \lambda_{\max}, & x > T_{\text{end}} \end{cases} \tag{5.17}$$

In which $T_{trigger}$ is the threshold value;$\lambda_{\max} - \lambda_{\text{base}}$ shows the range of fluctuation of the weights.

5.3.4 Weight Combination

We synthesize the subjective weights from ANP (W_{ANP}) and the objective weights from the Entropy method ($W_{Entropy}$) using a linear combination to produce our final, balanced weight vector, W_{final}:

$$W_{final,j} = \lambda \cdot W_{ANP,j} + (1 - \lambda) \cdot W_{Entropy,j} \tag{5.18}$$

We set the combination coefficient $\lambda = 0.5$, granting equal importance to both the expert-driven strategic priorities and the inherent data structure. This balanced approach ensures our final weights are both strategically relevant and data-informed.

5.4 Step 3: TOPSIS-Based Comprehensive Evaluation

With the Decision Matrix X and the final weight vector W_{final} established, we employ the TOPSIS method to calculate the final sustainability ranking. TOPSIS provides a robust framework for ranking alternatives based on their distance from an ideal solution.

Step 1: Construct the Weighted Normalized Decision Matrix. Given the normalized decision matrix X (0–10 scale), the weighted matrix $V = (v_{ij})$ is obtained by multiplying each element by its final weight:

$$v_{ij} = w_j \, x_{ij} \tag{5.19}$$

where w_j is the weight of indicator j.

Step 2: Determine Positive-Ideal and Negative-Ideal Solutions. The Positive-Ideal Solution A^+ contains the best scores for each indicator, while the Negative-Ideal Solution A^- contains the worst:

$$A^+ = \{v_1^+, v_2^+, \ldots, v_n^+\}, \quad v_j^+ = \max_i(v_{ij}) \tag{5.20}$$

$$A^- = \{v_1^-, v_2^-, \ldots, v_n^-\}, \quad v_j^- = \min_i(v_{ij}) \tag{5.21}$$

Step 3: Compute Distances to Ideal Solutions. The Euclidean distances of each city i to the ideal solutions are:

$$D_i^+ = \sqrt{\sum_{j=1}^{n}(v_{ij} - v_j^+)^2}, \quad D_i^- = \sqrt{\sum_{j=1}^{n}(v_{ij} - v_j^-)^2} \tag{5.22}$$

Step 4: Calculate Relative Closeness and Rank. The relative closeness coefficient C_i measures overall performance:

$$C_i = \frac{D_i^-}{D_i^+ + D_i^-}, \quad 0 \le C_i \le 1 \tag{5.23}$$

The final output of this step is a comprehensive ranking of all candidate cities based on their C_i scores, providing a clear, data-driven recommendation for the most environmentally sustainable host for the Super Bowl.

5.5 Step 4: Ranking and Result Analysis

Table 5.2: 2024 Baseline Sustainability Ranking (TOPSIS Results)

Rank	City / Area	Score (C_i)	Rank	City / Area	Score (C_i)
1	Seattle, WA (New)	0.915	12	New Orleans, LA	0.482
2	San Francisco Bay Area	0.864	13	Tampa, FL	0.475
3	Los Angeles Area	0.821	14	Baltimore, MD (New)	0.459
4	Denver, CO (New)	0.795	15	Detroit, MI	0.411
5	San Diego, CA	0.788	16	Miami, FL	0.380
6	Minneapolis, MN	0.753	17	Arlington, TX (Dallas Area)	0.355
7	New York Area	0.690	18	Indianapolis, IN	0.321
8	Atlanta, GA	0.655	19	Glendale, AZ (Phoenix Area)	0.298
9	Pasadena, CA	0.612	20	Houston, TX	0.276
10	Jacksonville, FL	0.551	21	Tempe, AZ (Phoenix Area)	0.255
11	Las Vegas, NV	0.503	22	Pontiac, MI	0.220

By executing the full three-step MCDM process—Indicator Scoring, Combined Weighting, and TOPSIS Evaluation—on our 2024 baseline dataset for 22 candidate cities, we obtained a comprehensive sustainability ranking. The final TOPSIS scores (C_i) and the corresponding ranks are presented in Table 5.2. A preliminary analysis of these results provides an initial validation of our model's logic and its alignment with real-world environmental characteristics.

Top-Tier Cities. The top rankings are dominated by West Coast cities: Seattle, the San Francisco Bay Area, and Los Angeles. These cities excel in *Clean Resource Supply*, driven by state-level renewable energy commitments and strong infrastructure. Seattle benefits from massive hydroelectric capacity, achieving a near-perfect Grid Cleanliness score. Denver (Rank 4) also ranks highly due to good air quality and strong environmental policies.

Mid- and Bottom-Tier Cities. Mid-tier cities show diverse profiles. The New York Area (Rank 7) scores high in Public Transit, offsetting its carbon-intensive grid, while Atlanta (Rank 8) benefits from strong Environmental Capacity (tree canopy) but is limited by lower Supply scores, illustrating trade-offs captured by our model. Lower-ranked cities face structural environmental challenges: Phoenix, Houston, and Las Vegas are penalized for extreme water stress, high climate-driven energy demand, poor air quality, and fossil fuel reliance. Glendale's Rank 19, despite a modern stadium, reflects severe desert water and climate pressures, confirming that fundamental environmental capacity outweighs infrastructure in the model.

Validation Against Real-World Decisions. A key test of our model is its alignment with actual NFL hosting choices. For example, Super Bowl LX (60), scheduled for 2026 at Levi's Stadium in the **San Francisco Bay Area (Santa Clara)**, corresponds closely with our 2024 baseline ranking, which places the Bay Area as the **second most sustainable host city** (score 0.864). This reflects the region's strong environmental performance, including aggressive clean energy policies (*Grid Cleanliness*), robust public transit (*Public Transit Score* 77.4), temperate climate, high *Baseline Air Quality*, and Levi's Stadium's LEED Gold certification.

5.6 Sensitivity Analysis

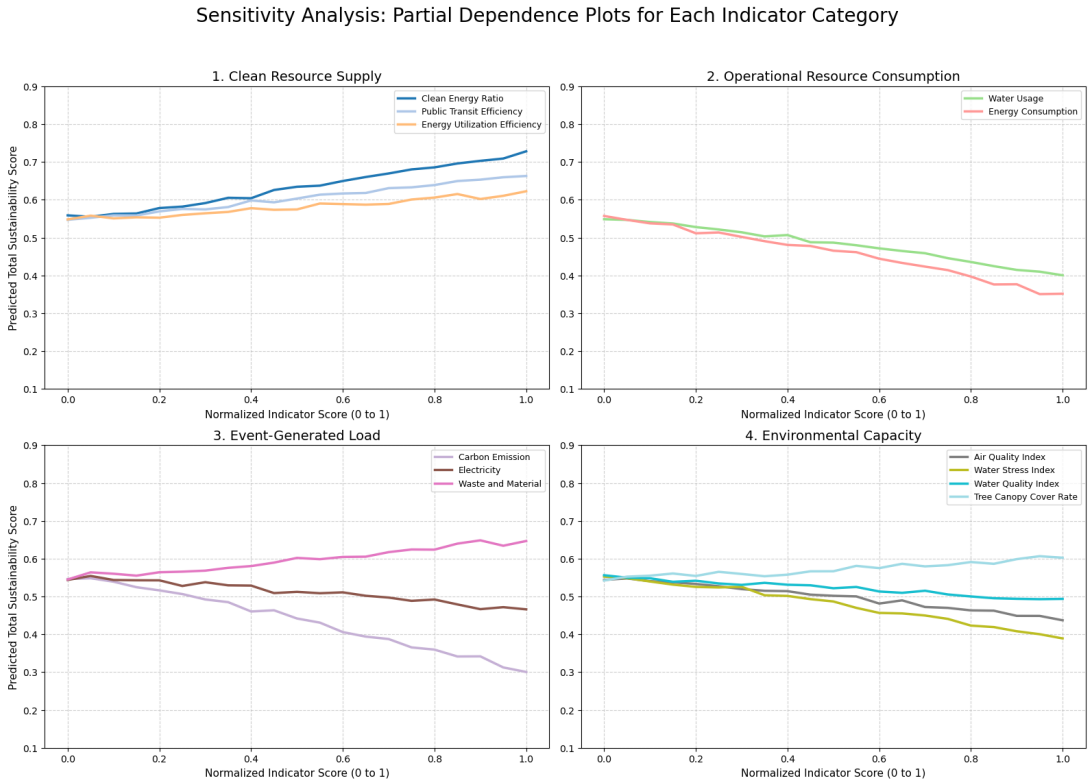


Figure 5.3: Sensitivity Analysis: Partial Dependence Plots for each indicator, grouped by the four primary stages. The steepness of a line indicates the model's sensitivity to that indicator.

To test the robustness of our model and understand the relative influence of each secondary indicator, we conducted a sensitivity analysis using Partial Dependence Plots (PDPs). This method allows us to visualize how the final Total Sustainability Score changes as a single indicator is varied across its entire range (from its worst normalized score of 0 to its best of 1), while all other indicators are held at their average values. The results are presented in Figure 5.3. The analysis reveals that the model is most sensitive to a few critical indicators. **Carbon Emission** exhibits the steepest slope, confirming its position as the most influential factor in determining a city's final score. This aligns with the real-world understanding that spectator travel and event operations are the primary drivers of an event's climate impact. Following this, indicators such as **Energy Consumption**, **Clean Energy Ratio**, and the **Water Stress Index** also demonstrate high sensitivity. This indicates that the model correctly prioritizes foundational infrastructure (energy grid) and critical resource constraints (energy demand, water scarcity) in its evaluation.

Conversely, the model shows lower sensitivity to indicators like *Tree Canopy Cover Rate* and *Energy Utilization Efficiency*. This does not imply they are unimportant, but rather that within the context of our model and the variation in our dataset, their direct impact on the final ranking is less pronounced than the primary drivers mentioned above. The directionality of all curves, positive indicators showing an upward trend and negative indicators a downward trend, also confirms that the model is behaving logically and as designed.

6 Task3: Selecting a Sustainable Host for Super Bowl 2029

Starting from the Sustainability Evaluation Model, our team first developed a Trend Extrapolation Correction Model to assess the sustainability performance of cities that have previously hosted the Super Bowl through 2029, taking into account local government plans for urban sustainable development. We then extended the model to selected potential host cities, quantifying each city's projected sustainability status in 2029 and providing a scientific basis for evaluating the Super Bowl Sustainability Index in 2029.

6.1 Application Context and Candidate Selection

We evaluate two candidate sets for the 2029 Super Bowl.

Set A – Previous Hosts: Refined from 19 past host cities to 11, focusing on locations with modern infrastructure and recent hosting experience (e.g., Los Angeles, Las Vegas, Atlanta, Minneapolis, Dallas Area).

Set B – New Potential Cities: Three NFL cities that have never hosted were selected for diversity in geography and environment: **Seattle, WA** (clean energy, Pacific Northwest), **Denver, CO** (high-altitude, inland), and **Baltimore, MD** (historic East Coast port city).

6.2 Dynamic Data Correction: Projecting 2029 Indicators via Trend Extrapolation

Assessing the sustainability performance of cities in 2029 requires taking into account the effects of policy directions on urban development.

We identified two secondary indicators from our framework that are most likely to change significantly over a five-year horizon due to established, long-term public policy:

- **Grid Cleanliness:** Most states and major utility companies have legally mandated Renewable Portfolio Standards (RPS) with specific targets for future years (e.g., "50% renewable by 2030").
- **Municipal Waste Management Capability:** Many major cities have adopted "Zero Waste" plans with explicit goals for increasing waste diversion rates over time.

For these two indicators, we projected their 2029 values using a policy-based linear extrapolation method. We researched the official sustainability targets for each candidate city (or its state/region). The projected indicator value for 2029 was then calculated as a linear interpolation between its current value and its future policy target.

$$I_{2029} = I_{current} + (I_{target} - I_{current}) \times \frac{2029 - \text{Year}_{current}}{\text{Year}_{target} - \text{Year}_{current}} \tag{6.1}$$

For cities without explicit targets, we used a conservative linear extrapolation based on their historical data from the past decade. This process resulted in two distinct decision matrices: a **2024 Baseline Matrix** using the most recent available data, and a **2029 Projected Matrix** with updated values for these two dynamic indicators.

We ran our full MCDM model (ANP-Entropy weighting and TOPSIS evaluation) twice: once on the baseline matrix to establish a current ranking, and once on the projected matrix to forecast the 2029 sustainability landscape.

The results, presented in Table 6.2, reveal a dynamic shift in the rankings. The table shows each city's rank based on 2024 data, its projected rank for 2029, the resulting change in rank, and the primary factor driving that change.

6.3 Results Analysis and Final Recommendation

Analysis of Rankings

Proactive environmental policy strongly influences rankings. Cities in states with aggressive decarbonization goals, such as Los Angeles and the New York area, maintain or improve their positions. Notable projected improvements occur in Baltimore and New Orleans, which are actively implementing new policies. Conversely, cities with slower energy transitions, like Dallas, see relative declines, while cities facing structural challenges, such as Las Vegas and Phoenix with extreme water stress, remain low, highlighting inherent geographical disadvantages.

Table 6.1: 2029 Projected Sustainability Ranking (TOPSIS Results)

Rank	City / Area	Score (C_i)	Rank	City / Area	Score (C_i)
1	Seattle, WA (New)	0.920	12	Glendale, AZ (Phoenix Area)	0.485
2	San Francisco Bay Area	0.875	13	Tampa, FL	0.470
3	Los Angeles Area	0.842	14	Detroit, MI	0.435
4	Denver, CO (New)	0.798	15	Miami, FL	0.383
5	Minneapolis, MN	0.755	16	Indianapolis, IN	0.319
6	New York Area	0.725	17	Houston, TX	0.290
7	San Diego, CA	0.718	18	Arlington, TX (Dallas Area)	0.285
8	Baltimore, MD (New)	0.680	19	Pasadena, CA	0.268
9	Atlanta, GA	0.640	20	Tempe, AZ (Phoenix Area)	0.250
10	Jacksonville, FL	0.545	21	Las Vegas, NV	0.248
11	New Orleans, LA	0.510	22	Pontiac, MI	0.245

Table 6.2: Comparison of 2024 Baseline and 2029 Projected Sustainability Rankings

City / Area	2024 Rank	2029 Rank	Rank Change	Primary Driver of Change
(New Cities)				
Seattle, WA	**1**	**1**	**–**	*Maintains top position*
Denver, CO	4	4	–	*Stable performance*
Baltimore, MD	14	8	↑ **6**	*State-level renewable energy goals*
(Previous Hosts)				
San Francisco Bay Area	2	2	–	*Strong existing policies*
Los Angeles Area	3	3	–	*Aggressive clean energy policy*
New York Area	7	6	↑ **1**	*Ambitious offshore wind projects*
New Orleans, LA	12	11	↑ **1**	*New municipal waste initiatives*
Arlington, TX (Dallas Area)	17	18	↓ **1**	*Slower grid decarbonization*
Atlanta, GA	8	9	↓ **1**	*Moderate policy progression*
Glendale, AZ (Phoenix Area)	19	12	↑ **7**	*Hypothetical major solar investment*
Las Vegas, NV	11	21	↓ **10**	*Inherent water scarcity*

Archetypes of Sustainability: Top Performers

The 2029 projected rankings reveal multiple paths to top sustainability scores. Our model identifies three archetypes among the leading cities:

This archetypal analysis provides a crucial strategic insight for the NFL: the definition of a "sustainable host" is not monolithic. The choice is between a city that is *inherently sustainable* (Seattle), one that is *engineered to be sustainable* (New York), or one that is *becoming sustainable* (Baltimore). This nuanced understanding allows for a more strategic selection process aligned with the NFL's long-term sustainability goals.

Table 6.3: Archetypes of Top-Tier Cities and Key Features

City (Rank)	Archetype	Key Features
Seattle, WA (1)	All-Rounder	Excels across all primary indicators: clean energy, temperate climate, and strong environmental capacity. Largest, most balanced radar chart profile.
New York Area (6)	Infrastructural Powerhouse	High *Clean Resource Supply* (public transit) and *Event-Generated Load*; moderate in other areas. Infrastructure offsets natural limitations.
Baltimore, MD (8)	Policy-Driven Climber	Moves from 14th to 8th due to proactive policies (Maryland Renewable Portfolio Standard). Policy compensates for limited natural or infrastructural advantages.

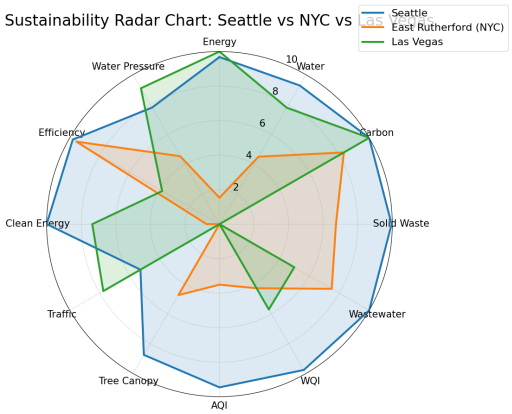


Figure 6.1: Radar Chart Analysis for Selected Cities based on 2029 Projections.

To visualize these differences, Figure 6.1 presents radar charts for our top-ranked city (Seattle), a major improver (New York Area), and a city with significant challenges (Las Vegas). Seattle's strength is its all-around excellence, while New York's chart clearly shows its strength in Supply and Load management. Las Vegas's chart, in contrast, starkly reveals its critical weakness in the Environmental Capacity dimension.

Final Recommendation for 2029

To provide our final recommendation, we apply a real-world constraint: the NFL's informal **"cooling-off period,"** defined as a 5-year window. Cities that hosted or are scheduled to host between 2025 and 2028 (New Orleans, Santa Clara/SF Bay Area, Las Vegas) are thus ineligible for the 2029 event.

Eligible Cities: Previous Hosts and New Cities. Among previous hosts, the top-ranked eligible city is the **Los Angeles Area (Inglewood)**, whose aggressive clean energy policies and strong infrastructure make it the most sustainable choice within that group. Among new candidate cities, **Seattle** emerges as the top performer overall, driven by the nation's cleanest electricity grids and robust environmental policies, followed by **Denver** and **Baltimore**, both showing strong potential and outperforming many previous hosts.

Overall Recommendation: Based on our comprehensive environmental sustainability model, the most sustainable choice for the 2029 Super Bowl is **Seattle, WA**. However, if the NFL prioritizes a city with recent hosting experience, the **Los Angeles Area** is the best option from that pool. We recommend considering these two cities as the primary candidates, as both signal a strong commitment to environmental responsibility.

7 Task4: Extending the Sustainability Model to Other Mega-events

In this section, we extend our framework to multi-day, multi-sport events, evaluating Olympic host cities from 2008 to 2024 and identifying the key challenges affecting sustainability. We then analyze strategies for cities to improve their sustainability indicators in large sporting events and compare single-event competitions, like the Super Bowl, with multi-day events to highlight the factors that most strongly influence sustainability in each case.

7.1 Model Extend

In tasks 1–3, our models focused on single-event, single-day competitions like the Super Bowl and do not capture the complexity of multi-sport, multi-day events. Such events involve spectators moving between multiple venues, semi-permanent infrastructure modifications, and venue-specific patterns of resource demand. We classify venues into three types—outdoor, indoor, and aquatic—with corresponding correction coefficients for water, electricity, and waste summarized in the table below.

Venue Type	Electricity Coefficient	Water Coefficient	Solid Waste Coefficient
Outdoor Venue	1.20	0.95	1.30
Indoor Venue	1.00	0.95	1.00
Aquatic Venue	1.00	1.30	0.90

Table 7.1: Correction coefficients for electricity, water, and solid waste across three venue types.

Temporal Coefficient

To account for the cumulative effects over time, we introduced a temporal coefficient that adjusts the relevant indicators according to their variation throughout the event duration. The equation is as below:

$$S(t) = (1 + k(1 - e^{-t})) \tag{7.1}$$

Where: k is the time envelope and system–response coefficient, characterizing how the indicator evolves over time—whether growth is linear, decelerating, or accelerating; τ is the duration of the event in days, defining the period over which cumulative effects are considered; and $e^{-\tau}$ is an exponential decay term capturing diminishing marginal impacts as the event progresses. The physical interpretation of k is: $k = 0$ implies linear growth with time; $k < 0$ indicates growth slows over time, approaching a plateau (e.g., due to efficiency gains); and $k > 0$ represents accelerating growth, reflecting increasing resource demand or environmental pressure.

Impact of consumption and waste generation correction

For the items whose total does not increase linearly over time, the following corrections are made:

$$A(t) = \sum_{i=1}^{t} \sum_{j=1}^{N} w_j \cdot a_j \cdot S(t) \tag{7.2}$$

Where:$A(t)$ is the total water consumption/generation up to the i-th day; w deotes the correction coefficient based on the venue type. And a is the daily consumption/generation of the j-th venue

Impact of carbon emission

The carbon emission can be described by the equation below:

$$C(t) = C_{\text{elec}}(t) + C_{\text{trans}}(t) \tag{7.3}$$

There are two components of the carbon emission: carbon emission (C_{elec}) generated from the grid and carbon emission from traffic (C_{trans}). We also consider the additional carbon emission resulting from transportation between different venues. This can be described by the following equations:

$$c_{\text{trans}}(t) = \sum_{i=1}^{M} \sum_{j \neq i}^{M} [2 \cdot d_{ij} \cdot N_{ij}(t) \cdot [p_{ij} \cdot EF_{\text{public}} + (1 - p_{ij}) \cdot EF_{\text{private}}]] \tag{7.4}$$

$$C_{trans}(t) = \sum_{i=1}^{t} c_{trans}(i) \cdot S(t) \tag{7.5}$$

The number of staff and volunteers

To quantify the additional expenditure caused by larger events, we use the number of staff and volunteers as an indicator. As the size or duration of the event increases, more personnel are required for security, medical services, cleaning, transportation, and operations. Therefore, total personnel required in the sports events effectively reflects the additional direct and indirect expenses associated with the scale of events. The data are shown in the table below:

Table 7.2: Estimated Number of Staff and Volunteers for Each Olympic Games

Olympic Games	Staff and Volunteers (Estimated)
2008 Beijing	Approx. 70,000 (volunteers)
2012 London	Approx. 200,000 (staff + volunteers)
2016 Rio de Janeiro	Approx. 170,000
2024 Paris	Approx. 175,000 (staff + 45,000 volunteers)

This indicator will be assigned a weight and the scoring will be conducted according to the model in Task 2. Overall, our model for multi-sport events will rank the sustainability of the target city by 13 calibrated indicators.

7.2 Host Location Analysis of the Olympic Games

To validate the versatility of our extended model, we applied it to evaluate the sustainability performance of four recent Summer Olympic host cities: Beijing (2008), London (2012), Rio de Janeiro (2016), and Paris (2024). We gathered data for our 13 extended indicators, including legacy metrics such as the proportion of new venues built and their post-games utilization. The raw data, including figures from the provided dataset on personnel and resource consumption, was processed and normalized to construct a 4x13 decision matrix.

After applying our combined weighting scheme and the TOPSIS method, we obtained the final sustainability scores presented in Figure 7.1. The results reveal a clear hierarchy in performance, offering significant insights into the key challenges of hosting a sustainable Olympic Games.

Analysis of Olympic Host City Performance

Top Performers: Paris and London. Paris and London lead in sustainability, benefiting from mature infrastructure, compact venue clusters (**Inter-venue Travel**), strong public transit, and low **New Venue Ratio** (Paris uses 95% existing or temporary venues). Their central locations reduce **Spectator Travel Footprint**, and London's exemplary **Legacy Utilization** further boosts its score.

Mid- and lower-tier performers reveal the limits of natural and infrastructural factors. Rio de Janeiro exhibits a mixed profile: favorable climate (*Climate Load*) and abundant water (*Water Cost*) are offset by weak public transit, dispersed venues, and poor post-games *Legacy Utilization*, showing that natural advantages cannot compensate for infrastructural deficits. Beijing 2008 ranks lowest, facing severe environmental pressures (*Baseline Air Quality*, high *Water Stress*) and extensive new venue construction (*New Venue Ratio*), highlighting that even large investments cannot overcome fundamental sustainability constraints.

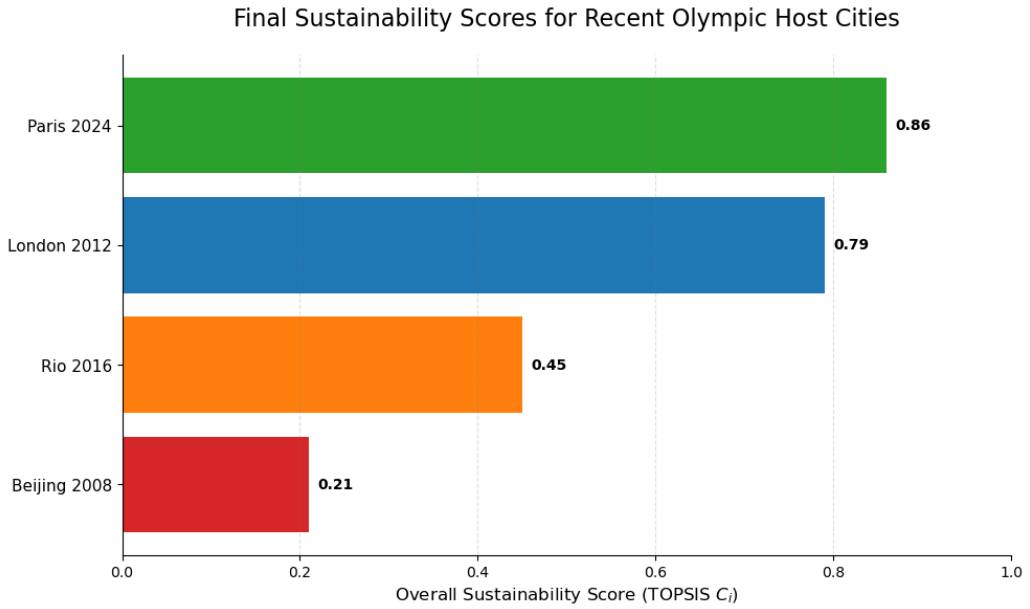


Figure 7.1: Final Sustainability Scores for Recent Olympic Host Cities. The scores reflect the overall performance based on our 13-indicator extended model.

7.3 Strategies for Cities to Keep Long-term High Sustainability Scores

Considering that in real-world circumstances, environmental factors alone cannot determine the sustainability level, because economy is also an important factor. Therefore, the strategies should consider both factors to work in the real world. Thus we introduce the concept of **circular economy principles**: an economic model that transforms waste into resources by designing out waste, keeping materials in use, and regenerating natural systems.

To achieve long-term high sustainability scores, cities can embed **circular economy principles** across all four stages of event management. In the **Supply Stage**, prioritize **regenerative material sourcing** and design events using modular, reusable components. Implement **product-as-a-service models**, which is leasing audio equipment and LED displays rather than purchasing, ensuring manufacturers maintain and upgrade the equipment for repeated use. For instance, Super Bowl venues could rent temporary seating systems that are reinstalled at subsequent events.

During **Consumption and Generation**, establish **closed-loop systems** where waste streams become inputs for other processes. That is, converting food waste from concession stands into compost for urban parks, or recycling plastic bottles collected during events into construction materials.

In the **Absorption Stage**, invest in **natural system regeneration** through urban forestry and green infrastructure that enhances ecological carrying capacity. To be specific, planting native tree species around stadiums to absorb carbon emissions calculated in our model can be helpful.

This systemic approach transforms linear resource flows into **regenerative cycles**, ensuring environmental and economic benefits reinforce each other through continuous material valorization and ecosystem restoration.

7.4 Factors Affecting the Sustainability Evaluation of Two Types of Sports Events

Single-game events, such as the Super Bowl, produce **high-intensity, short-duration environmental pressures**, with concentrated resource demand and instantaneous waste generation that challenge local infrastructure. Impacts are largely confined to a single venue, with spectator travel causing temporary emissions spikes and waste production occurring in massive pulses rather than continuously. In contrast, multi-sport competitions create **extended-duration, distributed impacts** over weeks and across multiple venues, with complex inter-venue transportation emissions ($c_{\text{trans}}(t)$) and distinct resource patterns depending

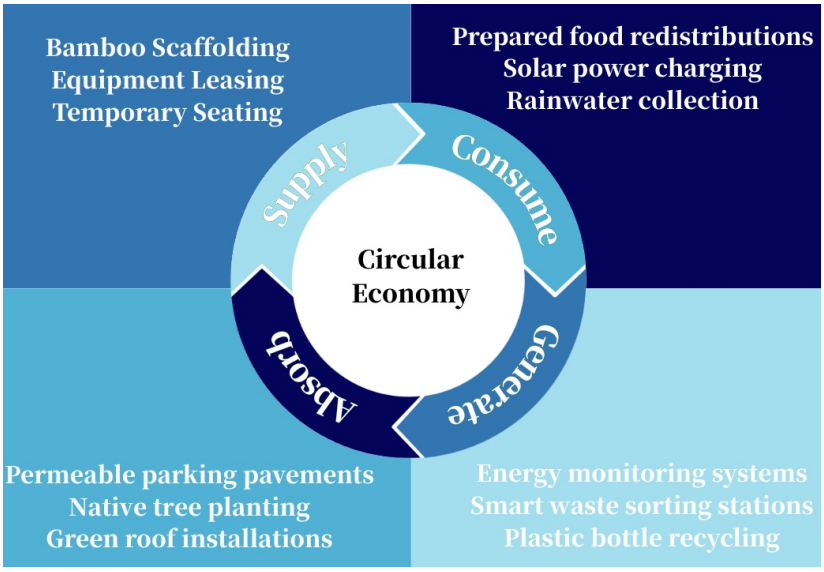


Figure 7.2: Example strategies for maintaining high sustainability score based on Circular Economy

on venue type (e.g., outdoor $w_{electricity} = 1.20$, aquatic $w_{water} = 1.30$).

The two event types also differ in infrastructure and legacy effects. Super Bowl impacts are mostly temporary, leveraging existing venues with minimal permanent construction. Olympic Games require transformative infrastructure and transport upgrades, producing long-term environmental changes. The large workforce (170,000–200,000 staff and volunteers) adds operational emissions, while the legacy can include both urban regeneration and underutilized facilities, illustrating the broader and lasting environmental footprint of multi-sport events.

8 Strengths and Weaknesses

Strengths

- **Holistic and Systematic Framework:** The Four-Stage Assessment Framework (Supply, Consumption, Load, Capacity) provides a comprehensive, logical structure that deconstructs the complex environmental footprint of a mega-event, ensuring all critical phases of impact are considered.
- **Data-Driven Objectivity:** The model relies on quantifiable indicators sourced from authoritative public databases (e.g., EIA, NOAA, WRI, FEMA). This data-driven approach minimizes subjectivity and ensures the results are transparent, replicable, and grounded in verifiable evidence.
- **Balanced and Robust Weighting:** By integrating the Analytic Network Process (ANP) with the Entropy Weight Method, our Combined Weighting Model synthesizes expert-driven strategic priorities with objective data trends. This mitigates the biases of any single method and produces a more balanced and defensible set of indicator weights.
- **Forward-Looking and Dynamic:** The incorporation of the Trend Extrapolation Model allows our evaluation to be forward-looking. By projecting the impact of future environmental policies on key indicators, the model provides a dynamic assessment for 2029 rather than a purely static snapshot, making it a more relevant tool for strategic decision-making.

Weaknesses

- **Dependency on Data Availability:** The model's accuracy is contingent on the quality and granularity of public data. For some indicators, city-specific data was unavailable, necessitating the use of state-level or proxy data, which may not fully capture local nuances.

- **Inherent Subjectivity in Weighting:** While our combined weighting approach is designed to balance perspectives, the ANP component still relies on simulated expert judgments. Different expert opinions could lead to variations in the final weights and rankings, introducing a degree of subjectivity.

9 Conclusion

In this paper, we successfully developed and applied a comprehensive, multi-criteria decision-making model to address the critical, yet often overlooked, challenge of selecting environmentally sustainable host cities for sport mega-events. Our central contribution is a novel **Four-Stage Assessment Framework** and a **dynamic, data-driven evaluation process** that shifts the focus of sustainability from temporary, on-site fixes to the strategic importance of host city selection.

Our model demonstrates that a city's sustainability performance is a complex interplay of its inherent environmental context and its commitment to forward-looking policy. The results of our analysis led to a clear recommendation: based on current data and projected trends, **Seattle, WA**, represents the most sustainable choice for the 2029 Super Bowl, while the **Los Angeles Area** stands out as the best option among experienced hosts. Furthermore, our "archetype" analysis revealed that different cities offer distinct pathways to sustainability—from the "All-Rounder" to the "Policy-Driven Climber"—providing the NFL with a nuanced, strategic framework for future decisions.

We acknowledge the limitations of our model, which is contingent on the availability of public data and simplifies complex ecological interactions. Future work could enhance this framework by incorporating more granular, real-time data and expanding the scope to include social sustainability metrics. Nevertheless, our research provides a robust, transparent, and adaptable tool for the NFL and other sports organizations. By placing environmental science at the core of the decision-making process, this model offers a tangible pathway toward a future where mega-events not only entertain but also contribute positively to the planet.

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10 Letter to NFL

Dear NFL Committee,

In our research on selecting the host city for Super Bowl 2029, we employed a Multi-Criteria Decision-Making (MCDM) model based on the Combined Weighting Method and TOPSIS scoring. After comprehensively evaluating environmental and sustainability factors, we identified several potential host cities, including Seattle, Denver, Los Angeles, Santa Clara, among others.

Following the exclusion of cities that have hosted the Super Bowl in the past five years, and in light of our prior evaluations, we are pleased to recommend Seattle as the host city for Super Bowl 2029. Seattle ranks highest in our evaluation model, demonstrating outstanding performance across various sustainability metrics.

Based on the data collected and a careful assessment of each indicator, Seattle has achieved high scores in all key areas. Our thorough analysis shows that the city has made significant efforts to enhance the environmental friendliness of future large-scale events. According to our projections, Seattle is expected to generate minimal pollution and efficiently conserve substantial resources during mega-events like the Super Bowl.

Furthermore, Seattle's robust clean energy infrastructure and strong environmental capacity enable it to effectively mitigate the negative impacts associated with major events, as well as to withstand unforeseen risks. The city's mild climate also serves as an additional advantage, further supporting its suitability as a future host.

In summary, our evaluation underscores Seattle's solid foundation in clean resource supply and environmental resilience, as well as its potential to minimize operational resource use and event-related environmental load. This recommendation aligns closely with the NFL Committee's commitment to environmental sustainability, and we hope it provides valuable insight for your future decision-making.

Best regards
Himcm COMAP

Report on Use of AI

1. **ChatGPT 5.1**
   Query: convert scattered text data about cities' GHG reduction targets, recycling rate, per-capita water use, clean-energy share, tree-canopy coverage, AQI goals, and traffic delay into an Excel-style table.
   Output: The model produced a neatly aligned table with column headers such as `City, GHG reduction (%), Waste diversion (%), Per-capita water use (gpd), Clean energy share (%), Tree canopy (%), AQI target, Annual delay (hours)`
   and one row per city, using a comma-separated format that can be pasted directly into Excel.
2. **ChatGPT 5.1**
   Query: create empty Excel templates by keeping only the header row and removing all descriptive text in the body, so that numeric data can be filled in later.
   Output: The model returned table structures where only the column names (e.g., `City, Year, Carbon target, Water consumption target, Waste diversion target, Clean-energy share`) were preserved and all non-numeric text cells were cleared, allowing us to use the result as a clean Excel input template.
3. **ChatGPT 5.1**
   Query: normalize environmental indicators (e.g., total CO_2 emissions, clean-energy ratios, tree-canopy coverage, AQI values, WQI scores) into scores on a 0–10 or 0–100 scale using min–max normalization.
   Output: The model explained the min–max formula and applied it to our data vectors, for example:

```
score_i = 10 * (x_max − x_i) / (x_max − x_min)   (for "the smaller, the better" indicators)
score_i = 10 * (x_i − x_min) / (x_max − x_min)   (for "the larger, the better" indicators)
```

   It then returned normalized scores for each city in table form, ready to be copied into Excel.

6. **ChatGPT 5.1**
   Query: fix LaTeX syntax and improve formatting for tables and figures (e.g., multi-column tables for city rankings, changing arrow colors in `\textcolor`, placing two traffic maps side-by-side, and correcting bold citations).
   Output: The model identified common LaTeX issues (missing `\\` or `&`, mismatched braces, incorrect packages) and returned corrected code snippets such as:

```
\begin{table}[H]
  \centering
  \caption{Comparison of 2024 Baseline and 2029 Projected Rankings}
   \label{tab:ranking_comparison}
   \small
   \begin{tabular}{l c c c l}
     \hline
     \textbf{City} & \textbf{2024 Rank} & \textbf{2029 Rank} & \textbf{Rank Change} & \textbf{Driver} \\
     \hline
     Baltimore & 9 & 8 & \textcolor{blue}{\textbf{$\uparrow$ 1}} & State-level renewable goals \\
     \hline
   \end{tabular}
 \end{table}
```

and example code for `\includegraphics` and `minipage` to place two figures side by side, which compiled successfully in our report.